A practical introduction to simulating complex trial designs

Introduction to simulation studies

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Motivation

Simulation studies are the **universally used tool** (and, in many instances, the only tool) for evaluations of novel statistical methods.

Simulations have been proven particularly useful when evaluating

- statistical methods for which **no closed-form analytical solutions** are available
- the properties of statistical methods with closed-form solutions **under range of conditions**

An assessment via simulations brings **a great deal of subjectivity**

- as it is based on the simulation conditions that are chosen by researchers, and
- the interpretation of findings can be equivocal due to methods' complexity.

Motivation for simulations in clinical trials

There is an increasing uptake of **more flexible and more efficient** (and hence more complicated) clinical trial designs.

There are two main classes:

- Adaptive designs
- Bayesian designs

Idea of adaptive designs

Modify an ongoing trial

by design

based on reviewing accrued data at interim

to enhance flexibility

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without undermining the study's integrity and validity[1].

[1] Dimairo et al (2020) The Adaptive designs CONSORT Extension (ACE) Statement: A checklist with explanation and elaboration guideline for reporting randomised trials that use an adaptive design. *BMJ - British Medical Journal*. 369

Idea of adaptive designs

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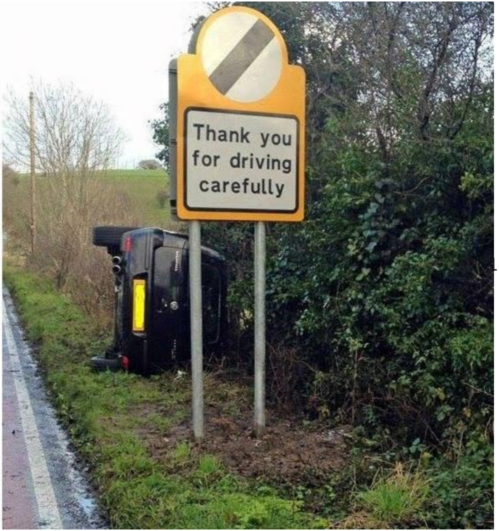

[1] Dimairo et al (2020) The Adaptive designs CONSORT Extension (ACE) Statement: A checklist with explanation and elaboration guideline for reporting randomised trials that use an adaptive design. *BMJ - British Medical Journal*. 369

Adaptive Designs[2]

- Benefits: "better" and/or faster decisions
- Challenges:
  - Operationally more complex
  - Care needed around false decision risk
  - Risk of bias
- For (complex) adaptations, such designs **typically require simulations** as closed-form solutions for type I error and power might not be available

[2] Pallmann et al. (2018) Adaptive designs in clinical trials: why use them, and how to run and report them. *BMC Medicine*. 16:29

Bayesian Designs

Prior Opinion → Data → Posterior Opinion

- Determining a prior
- Updating the prior with data
- Make conclusion based on full distribution of effect

\+ Can use all information available for decision making

\+ Provide information on a scale that stakeholders are often more familiar with

– Sample size estimation can be complex

– Prior input always present at final result

- Simulations are needed as there are **no closed-form solutions** for complex (realistic) models;
- Simulations are needed to understand **the impact of the priors**

Definition of a simulation study

- A **simulation study** consists of computer generated data sets (simulated trials)
- Each **simulated trial** consists of outcomes for hypothetical subjects given the set of conditions (simulation scenarios)
  - distributions of outcomes;
  - characteristics of study participants;
  - sample size;
  - mechanism of missingness;
  - recruitment rates
  - ...
- **Repeated realisations** of simulated trials are analysed and operating characteristics (OCs) are computed to describe trial properties
- **Multiple scenarios** studied to reach an overall conclusion of trial performance

Role of simulations for design of clinical trials

Main objectives:

- calculate sample size, type I error and power (and more)
- understand the behaviour under various design and scenario parameters and misspecification(s)

We also use the simulations to

- Check that method (and the code) works as intended
- Better understand statistical concepts
- Confirm mathematical results
- Compare methods head-to-head

(Un)popular opinion

Phase I trial design: Is 3+3 the best?[3]

> *The evidence from this review suggests that the 3+3 design identifies the recommended phase 2 dose and toxicities with an acceptable level of precision in some circumstances*

> *Novel trial designs demonstrating superiority over the 3+3 method in statistical simulations without corroborating clinical evidence are of theoretical value alone*

What comes first the simulations (chicken) or the practice (egg)?

Simulations should define what we will **actually do in practice**!

[3] Hansen AR, Graham DM, Pond GR, Siu LL. (2014) Phase 1 trial design: is 3+ 3 the best?. Cancer control. 21(3):200-208

Regulatory perspective - FDA

FDA guidelines on adaptive designs[4] and Interacting with the FDA on CIDs[5] highlight importance of simulations

> *For many Complex Innovative Designs (CIDs) the properties (e.g. type I error, power, sample size, etc.) cannot be derived analytically. Simulation studies are often required.*

- Sponsors are encouraged to engage with FDA early including discussions of a simulation plan
- **Special attention to**: type I error control (false positives), power, bias in estimation and confidence interval coverage
- Simulation is **one of the inputs** into that decision, not the only one.

[4] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adaptive-design-clinical-trials-drugs-and-biologics-guidance-industry

[5] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/interacting-fda-complex-innovative-trial-designs-drugs-and-biological-products

Regulatory perspective - ICH E20

ICH E20 adaptive designs for clinical trials[6]

> *If simulations are critical to understand operating characteristics of an adaptive design, the simulation study should be carefully planned, conducted, and reported. All relevant details pertinent to the planning of an adaptive trial design should be documented*

- **When:** where trial features cannot be analytically derived
- **Why:**
  - help ensure that after adaptations, estimates remain reliable
  - show that under the planned design, Type I error is controlled
  - show risks of false negatives under reasonable scenarios
- **Reporting:** Statistical analysis plan should include a description of the simulation plan

[6] https://www.ema.europa.eu/en/ich-e20-adaptive-designs-clinical-trials-scientific-guideline

ADEMP reporting framework[7]

- **A**ims
  Set out what question(s) the simulation study is designed to address
- **D**ata Generation
  Set out how the simulated data will be created
- **E**stimands
  What quantity the analysis is trying to estimate
- **M**ethods
  Set out how each simulated data set will be analysed to give an estimate of the estimand
- **P**erformance measures
  What is a good result?

ADEMP can enhance the validity and transparency of simulations

[7] Morris TP, White IR, Crowther MJ. (2019) Using simulation studies to evaluate statistical methods. Statistics in Medicine. 38(11):2074-2102.

OCTAVE framework[8]

- **O**bjective(s) of conducting a clinical trial simulation
  What are we trying to achieve
- **C**haracteristics of underlying factors
  All factors relevant to represent the (simulated) world. E.g. Endpoint distribution, recruitment rate,...
- **T**rial design(s)
  Design options to be evaluated
- **A**nalysis methods
  How will the generated data be analysed
- *V*aluation
  How will design and analysis strategies be compared against each other
- *E*vidence
  How will reproducibility be ensured and results reported

[8] Lee KM, Choodari-Oskooei B, Grayling MJ, Jacko P, Kimani PK, Mukherjee A, Pallmann P, Parke T, Robertson DS, Wang Z, Yap C, Jaki T. (2026) Clinical trial simulation: planning with the OCTAVE framework, implementation and validation principles. Statistics in Medicine.45(6-7):e70449.

Good practices

Statistical simulations should address important design/reporting issues:

- How to simulate data in a **realistic way**?
- How to simulate data to full **unveil the properties** of the design?
- How to ensure the **reproducibility** and transparency of the methods used for data generation and analyses?
- What **parameters and assumptions should be varied** across simulated scenarios?
- Which "**competing methods**" should be considered?
- How can the results be **interpreted**, without the risk of over-interpretation?

Good practices

It is often useful to **separate code** for the three tasks:

1. Function(s) to **generate data** sets
2. Function(s) to **analyse** the generated/observed data
3. Function(s) that calls 1. and 2 to **store/presents the results** in a well-structured outputs

Start small and build up slowly!

Good practices

Often we **vary several parameters at the same time** to understand interactions

To help navigate across huge (!) simulation studies with many different scenarios we can

- Plot the distribution across simulations
- Average across a design/model parameter to understand patterns
- Study "outliers"
- Explore individual datasets
- Include a comparator (reference method/benchmark)

Common operating characteristics

- Type I error
- Power
- Expected sample size
- Distribution of the realised sample sizes
- Properties of the treatment effect estimator
  - Bias
  - SE
  - MSE
- Width and coverage of the confidence/credible intervals
- Convergence
- Computational speed

Simulation (Monte Carlo) error

TABLE 6 Performance measures: definitions, estimates and Monte Carlo standard errors

Performance Measure	Definition	Estimate	Monte Carlo SE of Estimate
Bias	$E[\hat{\theta}] - \theta$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} \hat{\theta}_i - \theta$	$\sqrt{\frac{1}{n_{sim}(n_{sim}-1)} \sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \bar{\theta})^2}$
EmpSE	$\sqrt{\mathrm{Var}(\hat{\theta})}$	$\sqrt{\frac{1}{n_{sim}-1} \sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \bar{\theta})^2}$	$\frac{\widehat{\mathrm{EmpSE}}}{\sqrt{2(n_{sim}-1)}}$
Relative % increase in precision (B vs A) [a]	$100\left(\frac{\mathrm{Var}(\hat{\theta}_A)}{\mathrm{Var}(\hat{\theta}_B)} - 1\right)$	$100\left(\left(\frac{\widehat{\mathrm{EmpSE}}_A}{\widehat{\mathrm{EmpSE}}_B}\right)^2 - 1\right)$	$200\left(\frac{\widehat{\mathrm{EmpSE}}_A}{\widehat{\mathrm{EmpSE}}_B}\right)^2 \sqrt{\frac{1-\mathrm{Corr}(\hat{\theta}_A,\hat{\theta}_B)^2}{n_{sim}-1}}$
MSE	$E[(\hat{\theta} - \theta)^2]$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \theta)^2$	$\sqrt{\frac{\sum_{i=1}^{n_{sim}} \left[(\hat{\theta}_i - \theta)^2 - \widehat{\mathrm{MSE}}\right]^2}{n_{sim}(n_{sim}-1)}}$
Average ModSE[a]	$\sqrt{E[\widehat{\mathrm{Var}}(\hat{\theta})]}$	$\sqrt{\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} \widehat{\mathrm{Var}}(\hat{\theta}_i)}$	$\sqrt{\frac{\widehat{\mathrm{Var}}[\widehat{\mathrm{Var}}(\hat{\theta})]}{4n_{sim} \times \widehat{\mathrm{ModSE}}^2}}$ [b]
Relative % error in ModSE[a]	$100\left(\frac{\mathrm{ModSE}}{\mathrm{EmpSE}} - 1\right)$	$100\left(\frac{\widehat{\mathrm{ModSE}}}{\widehat{\mathrm{EmpSE}}} - 1\right)$	$100\left(\frac{\widehat{\mathrm{ModSE}}}{\widehat{\mathrm{EmpSE}}}\right) \sqrt{\frac{\widehat{\mathrm{Var}}[\widehat{\mathrm{Var}}(\hat{\theta})]}{4n_{sim} \times \widehat{\mathrm{ModSE}}^4} + \frac{1}{2(n-1)}}$ [b]
Coverage	$\Pr(\hat{\theta}_{low} \le \theta \le \hat{\theta}_{upp})$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} 1(\hat{\theta}_{low,i} \le \theta \le \hat{\theta}_{upp,i})$	$\sqrt{\frac{\widehat{\mathrm{Cover.}} \times (1 - \widehat{\mathrm{Cover.}})}{n_{sim}}}$
Bias-eliminated coverage	$\Pr(\hat{\theta}_{low} \le \bar{\theta} \le \hat{\theta}_{upp})$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} 1(\hat{\theta}_{low,i} \le \bar{\theta} \le \hat{\theta}_{upp,i})$	$\sqrt{\frac{\widehat{\mathrm{B\text{-}E\ Cover.}} \times (1 - \widehat{\mathrm{B\text{-}E\ Cover.}})}{n_{sim}}}$
Rejection % (power or type I error)	$\Pr(p_i \le \alpha)$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} 1(p_i \le \alpha)$	$\sqrt{\frac{\widehat{\mathrm{Power}} \times (1 - \widehat{\mathrm{Power}})}{n_{sim}}}$

[a] Monte Carlo SEs are approximate for *Relative % increase in precision*, *Average ModSE*, and *Relative % error in ModSE*.

[b] $\widehat{\mathrm{Var}}[\widehat{\mathrm{Var}}(\hat{\theta})] = \frac{1}{n_{sim}-1} \sum_{i=1}^{n_{sim}} \{\widehat{\mathrm{Var}}(\hat{\theta}_i) - \frac{1}{n_{sim}} \sum_{j=1}^{n_{sim}} \widehat{\mathrm{Var}}(\hat{\theta}_j)\}^2$.

ICH E20 explicitly mentions $\geq 100,000$ for the estimation of type I error.

Discussion

Simulation is a powerful (and accessible) tool to better understand properties of clinical trial designs and to make an informed decision.

Unfortunately, many simulations studies are not reproducible or/and their rationale is not reproducible.

In this course, we will illustrate how to construct meaningful, reproducible, simulation studies.